Example Vehicle Service Manual

Sample document for the service-manual chatbot demo. Fictional values. Not a real vehicle.

1. Maintenance schedule

Interval	Item	Page
Every 7,500 km	Engine oil and filter	12
Every 15,000 km	Air filter, tyre rotation	14
Every 30,000 km	Spark plugs	18
Every 60,000 km	Timing belt	24
Every 90,000 km	Brake fluid	31

Items marked with an asterisk are inspection only and are not replaced unless worn past the limit in section 4.

2. Engine specifications

Valve clearance, cold engine

Engine	Intake	Exhaust
1.8 L	0.18 to 0.22 mm	0.23 to 0.27 mm
2.0 L	0.20 to 0.24 mm	0.25 to 0.29 mm

Spark plug gap: 1.0 to 1.1 mm for both engines. Torque the plug to 18 Nm. Do not adjust a platinum plug; replace it.

Ignition timing: 12 degrees BTDC at 750 rpm with the service connector jumped.

3. Timing belt

Tension check: with the engine cold, press the belt midway between the camshaft and the water pump. Deflection should be 8 to 10 mm at 10 kgf. A belt that deflects more than 12 mm, or shows glazing or cracks across the teeth, is replaced rather than re-tensioned.

Replacement interval: 60,000 km or 60 months, whichever comes first. Replace the tensioner and the water pump at the same time; the labour overlaps and the parts do not survive a second interval.

Alignment: the camshaft mark lines up with the notch on the back cover, and the crankshaft mark with the pointer on the oil pump housing. Rotate the engine twice by hand and re-check before the covers go back on.

4. Brakes

Bleeding the brakes: the order is right rear, left rear, right front, left front. Keep the reservoir above the minimum line throughout; a reservoir run dry pulls air back into the master cylinder and the whole procedure restarts.

Attach a clear hose to the bleed screw and submerge the free end in clean fluid. Have an assistant hold the pedal down, open the screw a quarter turn, close it, then release. Repeat until no bubbles appear in the hose.

Fluid: DOT 3 or DOT 4. Do not mix with DOT 5.

Pad minimum thickness: 1.5 mm. Disc minimum thickness is stamped on the hub face; below it the disc is replaced, not machined.

5. Under-dash fuse box

Location: behind the panel below the steering column, on the driver's side. Release the two clips at the top edge and let the panel hang.

No.	Rating	Circuit
1	10 A	Instrument cluster
2	15 A	Fuel pump
3	7.5 A	Interior lamps
4	20 A	Wiper motor
5	10 A	Horn

The diagram is printed on the inside of the panel. A fuse that blows again immediately indicates a short in that circuit, not a failed fuse.

6. Torque values

Fastener	Torque
Spark plug	18 Nm
Oil drain plug	35 Nm
Wheel nut	110 Nm
Cylinder head bolt	68 Nm, then 90 degrees
Timing belt tensioner bolt	45 Nm

Values are for clean, lightly oiled threads unless stated otherwise. A cylinder head bolt that has been reused more than once is replaced.